

Akhil Gattu

Data Engineer

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Summary

- Data Engineer with 4+ years of expertise in designing, building, and optimizing scalable data solutions, proficient in SQL, Python, data warehousing, data modeling, and ETL/ELT pipelines.
- Optimizing Data Build Tool (DBT) projects in a Snowflake environment by implementing incremental models, leveraging partitioning, tuning query performance, and reducing query runtime and cloud data warehousing costs.
- Expert in implementing robust data pipelines across leading cloud platforms like AWS and leveraging cloud service toolkits (EC2, S3, Lambda, Glue, Athena, AWS Pipeline, Redshift) to ensure efficient data storage, processing, and flow.
- Proficient in Managing and optimizing DBT projects in a Snowflake environment by implementing incremental models, leveraging partitioning, and tuning query performance, reducing query runtime and cloud data warehousing costs.
- Implementing Azure DevOps for CI/CD pipelines and utilizing Databricks for Spark-based data processing, transformations, and machine learning model development.

Skills

Programming Language: Python, R, SQL

Big Data Ecosystem: Apache Spark, Apache Kafka, Apache Nessie, Hadoop, Hive, HDFS, Zookeeper

Cloud: AWS (EC2, S3, Redshift, Glue, Athena, AWS Pipeline, Lambda, Kinesis, EMR), Azure (SQL Databases, Blob Storage, Cosmos DB, Azure Data Factory, Azure DevOps, Databricks, Azure Stream Analytics)

Visualizations: Tableau, Power BI, Excel

Packages: NumPy, Pandas, Matplotlib, Seaborn, PySpark

ETL and Tools: Fivetran, SSIS, Data Pipelines, Data build tool (DBT), Apache Airflow

Version Control & Database: GitHub, Git, SQL Server, PostgreSQL, Cassandra, MySQL, Snowflake

Education

Master in Information Systems and Technology

University of North Texas, Denton, TX

Experience

USAA, TX

Jan 2023 – Present

Data Engineer

- Spearheaded End-To-End Data Engineering Initiatives utilizing Python, PySpark, And Distributed Systems (Airflow, Databricks, Redshift, Snowflake, AWS) To orchestrate Robust and Scalable Data Pipelines collaborating with Cross-Functional Teams.
- Automated data pipelines using AWS Data Pipeline and Serverless Workflows (SWF) to orchestrate data movement and processing tasks, reducing manual intervention by 30%.
- Executed end-to-end data engineering initiatives utilizing Python, PySpark, and distributed systems (Airflow, Databricks, AWS Redshift) to orchestrate scalable data pipelines collaborating with cross-functional teams.
- Deployed a scalable data warehouse on Amazon Redshift, optimizing query performance through advanced query tuning and indexing strategies.
- Built ETL (Extract, Transform, Load) jobs using Glue crawlers and Spark scripts, automating data ingestion from various sources into a data warehouse, saving 2 hours of manual workload per week.
- Exploited AWS Redshift Spectrum for cost-effective processing of large datasets stored in S3, leading to a 20% reduction in data warehousing costs
- Employed Spark SQL, DataFrames, and Data Build Tool (DBT) transformations to perform complex data transformations and improve processing efficiency.

Dixon Technology, India

Oct 2019 – Nov 2021

Data Engineer

- Leveraged Snowflake's cloud-based architecture to facilitate collaboration between data analysts and data scientists, leading to a 30% increase in data-driven project completion rate.
- Accelerated PySpark for large-scale data processing and analytics tasks within the Apache Spark ecosystem and utilized DataFrames and SQL functionalities for efficient data manipulation.
- Built ETL pipelines using Azure Data Factory to extract, transform, and load data from various sources into Azure data stores, automating data integration processes and reducing manual effort by 50%.
- Designed and implemented Azure Data Factory pipelines to automate data ingestion, transformation, and loading (ETL) processes, resulting in a 30% reduction in manual effort for data processing tasks.
- Accomplished interactive Power BI reports leveraging advanced DAX and Power Query modeling techniques to unlock deeper data insights and facilitate informed decision-making.
- Improved ETL workflows by rigorous testing and debugging of SQL scripts and Python code resulting in a 50% increase in data processing efficiency and ensuring seamless data integration with downstream systems.
- Created Apache Flink applications for processing high-volume streaming data, achieving a 25% improvement in processing speed compared to traditional batch processing methods.